

Instructor: Date:

Learner: Regional Branch:

Program: Center:

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Let's Start!

A Very Talented Mathematician.



Learning Outcome:

Identify information about the history of software and its characteristics in simple spoken and written situations.

Ask your partner:

- Let's go back to the early years of the 20th century. Do you think it was difficult to create a device that solved mathematical problems?
- What kind of modern computing device would you like to create?

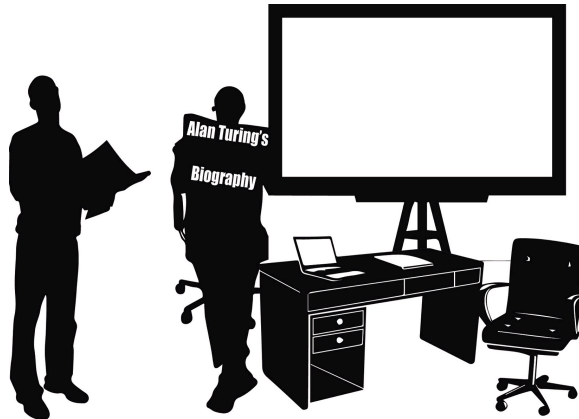
Can Do Statements:

- I can talk about the importance of Alan Turing for computing and software.
- I can give basic information about the Turing machine.
- I can discuss in general terms, how the Turing machine works.



Let's Explore!

Talking About Alan Turing and His Machine.



Instructions: Read the text below.

Ana: What are you reading?

Peter: This is Alan Turing's biography. I have to read it for my computer science class.

Ana: I see, and who is Alan Turing?

Peter: He was one of the many important characters in computing and software history.

Ana: Really? How so?

Peter: Well, hand me over your computer, I want to show you something. Look at his picture, he's one of the fathers of modern computing and I can tell you all about him. He basically conceived an idea called the Turing Machine; it involved the use of symbols in order to solve mathematical calculations. It is closely related to the development of the finite state machine.

Ana: Sounds difficult. I can understand only basic information on software. Please don't go into details.

Peter: Don't worry. Let me explain it to you in an easy way. The Turing Machine can write and read data on a tape. This machine uses symbols, similar to computer functions, in order to represent these data. For example: addition, subtraction, multiplication and some others.

Ana: Ok, you can go beyond with your explanation, I can understand better now.

Peter: Well, I guess we could say that Turing's theory helped to develop modern computers. Do you want to know, in general terms, how the machine works?

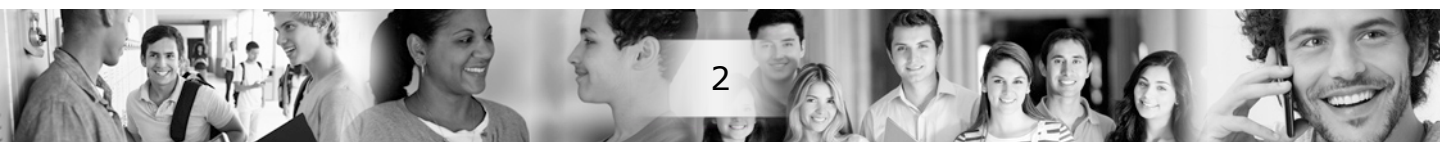
Ana: Sure. I have many questions about Turing's work.

Peter: Well, the machine's main function is to read and write data. It usually uses a paper tape to print cells with symbols on it. We could say that the Turing machine influenced many approaches in computing making a connection with grammar and language.

Ana: Really? Wow, you must have a very advanced level on software knowledge.

Peter: Not really, it's just a little bit of history. Oh, I forgot to tell you that Turing also contributed to the development of the Turing Test. Computer scientists use that test when working on artificial intelligence. That invention tries to answer one of the biggest questions of our times: Can computers think?

Ana: That sounds interesting.





Let's Practice!

Ana can do a recap* about the Turing Machine.

*A "recap" is a summary.

Instructions: Match column A with Column B.

COLUMN A

1. Can you tell me about the initial idea of the Turing Machine?
2. Can you describe the main function of the actual Turing Machine?
3. Can you give me some examples of computer functions?

COLUMN B

- a. It can read and write data on a tape.
- b. Yes, I can. Addition, subtraction, and multiplication are examples of that.
- c. I can say it involved a relationship between symbols and mathematical calculations

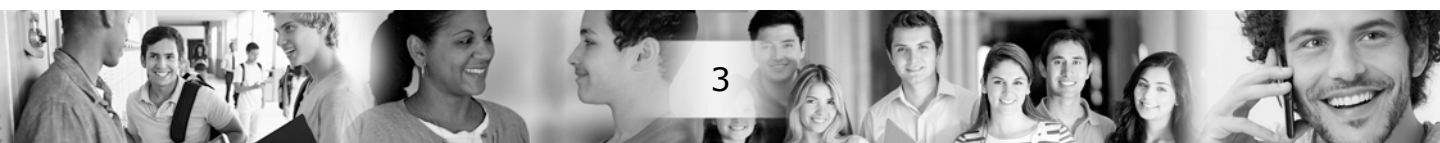


Let's Practice!

Let's go over the contributions of Alan Turing to computer science.

Instructions: Read the paragraph. Underline all the countable nouns that come with "many".

Many mathematicians think Alan Turing ideas were brilliant because he stated many theories based on computable numbers, and many of those theories included computer functions that were later used in electronic computers. Alan Turing was, in fact, an important person because many of his contributions were the cornerstone of computer programming. For example, the Turing Machine and the Turing Test are essential inventions for the world of computer science.





Extension Activity

Let's check inventions derived from the Turing Machine concept.

Instructions: Complete the following steps. Fill in the Self-Assessment.

- Look for further information about the devices and computer functions derived from the concept of the Turing Machine.
- Organize the information using some flashcards. On one side of the card, include a picture; on the other side, include basic information.
- Share your work with our classmates and instructor, and talk about it.

Self-Assessment

Yes No

- I deepen my knowledge about the Turing Machine.
- I express information through flashcards.
- I share information with my classmates and instructor.

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Target Words

- addition** (noun): the result of putting quantities together
e.g. The Turing Machine can write and read data on a tape, this data is represented in symbols which are equivalents to what we know nowadays as computer functions. For example: addition, subtraction, multiplication and some others.
- involve** (verb): to cover a specific matter
e.g. It involved the use of symbols in order to solve mathematical calculations.
- advanced** (adjective): a high level of something
e.g. Wow, you must have a very advanced level on software knowledge.
- hand over** (phrasal verb): to pass something to someone
e.g. Well, hand me over your computer. I want to show you something. Look at his picture, he's one of the fathers of modern computing.
- go beyond** (phrasal verb): to go further
e.g. Ok, you can go beyond with your explanation, I am starting to understand.
- tape** (noun): a long piece of material where information is written
e.g. The Turing Machine can write and read data on a tape, this data is represented in symbols which are equivalents to what we know nowadays as computer functions.



Learning Points

Verb Can to Talk About Ability

Modal verbs have different functions. The modal verb **can** is useful for expressing possibility, permission, or ability. Here, we'll focus on **can** for talking about **ability**.

When we talk about actions in general, we describe whether we are capable of doing those actions or not. We use **can** as a verb that shows that we're able to do something.

Follow the next structure to create statements with **can**:

Subject + can + verb (base form) + complement

Let's take a look at the following examples:

*I **can** talk about the history of Alan Turing.*
*She **can** describe the functions of the Turing Machine.*

Examples

- *The Turing machine can write and read data on a tape.*
- *I can understand only basic information, please don't go into detail.*
- *You can go beyond with your explanation.*

Quantifiers: Many

You use **many** as a quantifier of countable nouns. Remember that if a noun takes a plural form it is likely to be a countable noun.

Whenever we want to express that there is a large amount of something we can count, we use **many**. Please note that **many** is used for elements you can count (house-houses, dog-dogs) and much is used for elements that you cannot count (water, rice, love).

Many refers to a quantity above average. In order to use it in a sentence, you can follow the next structure:

Subject + verb + **many** + **countable noun** + complement

Example:

*I have **many questions** about Alan Turing. Is it true that he committed suicide? I read **many articles** about that.*





Learning Points

You can also use **many** after **there are**. For example:

There are many software developments nowadays.

Examples

- *This is Alan Turing's biography. I have to read many pages of this book for my computer science class.*
- *He was one of the many important characters for computing and software.*
- *I have many questions about Turing's work.*

How the Turing Machine Works

The **Turing Machine** was an important development of the mathematician Alan Turing. He invented it in 1936.

Some Important Facts

It's important to mention that the Turing Machine was an abstract model. It means that the nobody manufactured it. However, years after, people conceived it as a mathematical model of computation.

The basis of Turing's idea were symbols. Thanks to those symbols, mathematical calculations (adding, subtracting or multiplying) were possible. Later on, when people were developing electronic computers, the symbols of Turing's invent turned out to be computer functions. In a nutshell, the Turing Machine was the cornerstone of computer programming.

Some Information About How the Machine Works

We can define the Turing Machine as a mechanism that is capable of reading and writing data on a tape and it is divided into different states. Besides, It has an unlimited supply of tape that it can use as an input. First, it prints something; second, it moves to a new cell and then, it changes to a new state.





Learning Points

Let's imagine the tape as an endless belt full of little holes. That endless belt can be a synonym for the memory of a computer as we know it nowadays. In other words, the belt keeps all the information. Now, each hole in the belt represents a symbol and the machine can edit the symbol or erase it. The machine can also move the symbol to the right or to the left.

Take a look at the following image to get to know the functioning of the Turing Machine:

